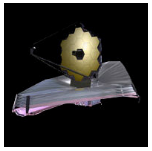

7201 - A deep look into the physics of the interstellar medium 360 Myr after the Big Bang

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
NIRSpec R2700 on GHZ2/GLASS-z12				
	1	G235H/F170LP GLASS-JWST	NIRSpec MultiObject Spectroscopy	(22) A2744_NIRSpec_R2700_G235_V7
	2	G395H/F290LP GLASS-JWST	NIRSpec MultiObject Spectroscopy	(23) A2744_NIRSpec_R2700_G395_V7

ABSTRACT

JWST observations have unexpectedly revealed a large number of distant luminous sources whose investigation is directing most observational and theoretical efforts. An outstanding example is provided by the MUV=-20.5 object GHZ2/GLASS-z12 at z=12.34, which has been observed with both NIRSpec PRISM and MIRI LRS, and with ALMA Band 6 and 8. The strength of its UV emission features is exceptional compared to the general high-redshift population and implies extreme ionization conditions. We will observe GHZ2/GLASS-z12 with the G235H/F170LP and G395H/F290LP configurations that cover all its key emission features, and whose resolving power enables to separate all density-sensitive line doublets (NIV, CIII, OII) and cleanly detect faint lines that are hardly measurable at low resolution. Coordinated ALMA Band 8 observations will reach ultra-deep limits on the [OIII]52m transition which provides a powerful density diagnostics in combination with the available [OIII]88m data. The proposed observations will provide firm constraints on the ISM of GHZ2/GLASS-z12 by measuring its electron density across 4 orders of magnitude and from low, to very high-ionization regions. The NIRSpec high-resolution observations will also robustly measure the HeII emission, confirm the highest-redshift detection of the Bowen fluorescence line OIII3133, and put tight constraints on very high-ionization features indicative of AGN emission. ALMA will reach deep limits on its dust continuum emission. This program will thus provide a thorough investigation of the ISM, ionization sources, abundances of an object that has the potential to become one of the best studied targets at very high-redshift.

OBSERVING DESCRIPTION

- JWST observations:

The program will observe the z=12.34 object GHZ2/GLASS-z12 with the goals of separately detecting the components of the density-sensitive line doublets and constrain faint emission features that are hardly measurable at low resolution.

The program comprises two NIRSpec MOS pointing on the GLASS-JWST NIRCam field, adopting the high-resolution configurations G235H /F170LP and G395H /F290LP.

We adopt readout pattern NRSIRS2, standard 3-shutter slits, a 3-point nodding and a 80-shutter dither along the dispersion direction to bridge the detector gap and obtain continuous coverage of the object. In order to maximise the exposure time, we constrained the MSA planning tool to observe at both dither positions the regions of all emission lines detected in the PRISM. As a result, the Aperture PA ranges needs to be restricted: the requested spectral coverage requires the G235H/F170LP and G395H/290LP configurations to be observed with two slightly different pointings both restricted at 25<PA<32.

Constrained source centering is chosen for an optimal positioning of the object in the slit. Remaining slits will be allocated to a rich set of filler targets, including robust high-redshift candidates, selected from the ultra-deep NIRCam imaging available on the GLASS-JWST ERS field.

The desired sensitivity (SNR>5-10 for the targeted emission lines and components) is reached with 6 exposures (3-nod x 2-dither positions), with 6 integrations of 15 groups per exposure in each of the gratings. The resulting total charged time on NIRSpec is 28.7 hours including overheads.

- ALMA observations:

We require deep [OIII]52 observations to infer a robust [OIII]52/[OIII]88 ratio of GHZ2/GLASS-z12 to accurately constrain its [OIII] electron density. The data will be used in combination with shallow [OIII]52 data from program 2023.A.00017.S and with the OIII]88 detection reported in Zavala et al. 2024. According to the public information from the ALMA Science Archive, the observations from program 2023.A.00017.S have a line sensitivity of 1.5mJy/beam in 10km/s channels. We request 2x deeper observations (0.7mJy/beam in 10km/s channels) that, according to the ALMA cycle-11 Observing Tool, requires 10.5hrs of observing time, including overheads. Since our main goal is to measure the line luminosity without significantly resolving the emission, we request observation with an angular resolution in the range of 0.15-0.7 arcsec (achieved with the configurations C-2 to C-5).

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(22)	A2744_NIRSpec_R2700_G235_V7	RA: 00 14 1.0287 (3.5042863d) Dec: -30 20 27.57 (-30.34099d) Equinox: J2000		
Comments: *Description=[]*				
(23)	A2744_NIRSpec_R2700_G395_V7	RA: 00 14 1.0287 (3.5042863d) Dec: -30 20 27.57 (-30.34099d) Equinox: J2000		
Comments: *Description=[]*				

Observation

Proposal 7201, Observation 1: G235H/F170LP GLASS-JWST Tue May 26 20:00:32 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(G235H/F170LP GLASS-JWST (Obs 1)) Warning (Form): Config c1 (#1) has 1 filler slits affected by failed closed shutters.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(22)	A2744_NIRSpec_R2700_G235_V7	RA: 00 14 1.0287 (3.5042863d) Dec: -30 20 27.57 (-30.34099d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	Filter: CLEAR; Readout: NRSRAPIDD6; 8 sources in 2 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	GHZs (10 sources)	FILLERS (13638 sources)	jwst-nirspec-g235h	1.5

Reference Stars

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
1	18687	3.518607	-30.340626	23.730100631713867	1	23158	3.505905	-30.315856	25.170499801635742
1	21278	3.502895	-30.311148	24.697599411010742	1	24115	3.480847	-30.319999	24.056299209594727
1	22284	3.482605	-30.318163	23.47450065612793	1	24340	3.484746	-30.315404	24.71139907836914
1	22892	3.497392	-30.324966	24.802000045776367	1	80846	3.528324	-30.320937	23.257400512695312

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G235H/F170LP)	c1	3 Shutter Slitlet	3.505017916666667 Degrees -30.340258333333338 Degrees	31.999631261450364			3	18	19957.602
2	1 (G235H/F170LP)	c2	3 Shutter Slitlet	3.4991760833333334 Degrees -30.337108888888906 Degrees	32.002591477736736			3	18	19957.602

Special Requirements	
	Aperture PA Range 25 to 32 Degrees (V3 246.42543030000002 to 253.42543030000002) MSA Scheduled Aperture PA 32.0000 to 32.0000 Degrees (V3 253.42543 to 253.42543)

Observation

Proposal 7201, Observation 2: G395H/F290LP GLASS-JWST Tue May 26 20:00:32 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 2:1) Warning (Form): The recommended value is 8 Reference Stars for this template.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(23)	A2744_NIRSpec_R2700_G395_V7	RA: 00 14 1.0287 (3.5042863d) Dec: -30 20 27.57 (-30.34099d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	Filter: CLEAR; Readout: NRSRAPIDD6; 6 sources in 3 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	GHZs-G395 (10 sources)	FILLERS-G395 (13638 sources)	jwst-nirspec-g395h	1.5

Reference Stars

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
1	20323	3.458245	-30.330468	23.642200469970703	1	22892	3.497392	-30.324966	24.802000045776367
1	21278	3.502895	-30.311148	24.697599411010742	1	23158	3.505905	-30.315856	25.170499801635742
1	21510	3.495597	-30.312172	25.13409996032715	1	82630	3.480011	-30.294938	24.505199432373047

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G395H/F290LP)	c1	3 Shutter Slitlet	3.4785095833333335 Degrees -30.32254999999998 Degrees	32.013070409051785			3	18	19957.602
2	1 (G395H/F290LP)	c2	3 Shutter Slitlet	3.472668875 Degrees -30.319398611111126 Degrees	32.01603011441317			3	18	19957.602

Special Requirements	Aperture PA Range 25 to 32 Degrees (V3 246.42543030000002 to 253.42543030000002) MSA Scheduled Aperture PA 32.0000 to 32.0000 Degrees (V3 253.42543 to 253.42543)